MilcharSec — Role-Based Cybersecurity Learning & Awareness Platform

1. Project Overview

Project Title: MilcharSec — Role-Based Cybersecurity Learning & Awareness Platform

Project Type: Web-Based Cybersecurity Education, Awareness, Assessment and Interactive Tools Platform

Target Audience: Non-technical employees, students, beginners, and individuals who want to improve their digital and workplace cybersecurity awareness.

What is MilcharSec?

MilcharSec is an interactive cybersecurity learning and awareness platform designed to help non-technical users understand, recognize, and respond to common cybersecurity threats.

Unlike a traditional e-learning website that only provides reading material and multiple-choice questions, MilcharSec combines:

- Structured cybersecurity learning modules
- Interactive lessons
- Knowledge quizzes
- Practical cybersecurity scenarios
- Simple security analysis tools
- User progress tracking
- Cybersecurity awareness scoring
- Personalized learning recommendations

The platform follows a continuous learning cycle:

Learn → Practice → Analyze → Quiz → Assess → Track Progress → Improve

The initial version of MilcharSec focuses on **Digital Security Basics and Cybersecurity Awareness for non-technical users**. In future versions, the platform can expand into specialized cybersecurity career paths such as SOC Analysis, Web Security, Digital Forensics, and Incident Response.

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2. Problem Statement

Cybersecurity incidents are not caused only by technical vulnerabilities. Human behavior is also a major security factor.

Many employees and non-technical users may:

- Use weak or reused passwords
- Fail to enable multi-factor authentication
- Click suspicious links
- Open unsafe attachments
- Fall victim to phishing or social engineering
- Share sensitive information incorrectly
- Use insecure public networks
- Ignore software updates
- Overshare personal or organizational information
- Fail to recognize a cybersecurity incident
- Not know how or when to report a security issue

Traditional cybersecurity training is often passive and theoretical. Users may read information or watch presentations but receive limited opportunities to apply what they have learned.

MilcharSec addresses this problem by combining cybersecurity education with interactive tools, practical exercises, quizzes, and progress tracking.

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3. Proposed Solution

MilcharSec provides a centralized platform where users can learn essential cybersecurity concepts and immediately apply their knowledge.

The platform allows users to:

1. Learn cybersecurity concepts through structured modules.
2. Interact with cybersecurity tools.
3. Analyze simulated suspicious emails, URLs, passwords, and security situations.
4. Complete quizzes and practical scenarios.
5. Receive scores based on their performance.
6. Track their learning progress.
7. Identify weak areas.
8. Receive recommendations about which module to study next.

The objective is not only to provide cybersecurity information but to help users develop practical awareness and safer digital behavior.

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4. Project Objectives

The main objectives of MilcharSec are:

- To improve cybersecurity awareness among non-technical users.
- To provide structured and understandable cybersecurity education.
- To teach users how to recognize common cyber threats.
- To provide interactive tools that support practical learning.
- To assess users through quizzes and scenario-based exercises.
- To track user learning progress and performance.
- To identify areas where a user requires improvement.
- To provide personalized learning recommendations.
- To create an engaging alternative to traditional cybersecurity awareness training.

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5. Target Users

The initial version of MilcharSec is designed primarily for non-technical users.

Primary Users

Employees

Employees who use email, computers, mobile phones, cloud applications, and organizational data but do not have a technical cybersecurity background.

Students

Students who want to understand the fundamentals of cybersecurity and safe digital behavior.

General Users

Individuals who want to improve their personal digital security and learn how to recognize common online threats.

Future Users

Future versions can introduce specialized role-based learning tracks for:

- SOC Analysts
- Security Analysts
- Web Security Analysts
- Digital Forensics Investigators
- Incident Responders
- Cybersecurity Interns

6. Core System Workflow

MilcharSec follows the following learning process:

User Registration/Login

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User Dashboard

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Select or Continue Learning Module

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Interactive Lesson Content

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Use Related Cybersecurity Tool

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Complete Practical Exercise or Scenario

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Attempt Quiz

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Receive Score and Feedback

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Update Progress and Awareness Score

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Identify Weak Areas

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Recommend Next Module

This creates a complete learning loop rather than simply displaying educational content.

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7. Cybersecurity Learning Modules

The initial MilcharSec platform contains **8 learning modules**.

Module 1 — Cybersecurity Fundamentals

This module introduces users to the basic concepts of cybersecurity.

Topics include:

- What is cybersecurity?
- Why cybersecurity is important
- Common cyber threats
- Malware basics
- Viruses and ransomware
- Social engineering
- Personal and organizational assets
- Confidentiality, Integrity and Availability
- Basic cybersecurity practices
- Human role in cybersecurity

Learning Outcome

After completing this module, users should understand the purpose of cybersecurity and recognize common digital threats.

Module 2 — Password & Account Security

This module focuses on protecting user accounts.

Topics include:

- Strong and weak passwords
- Password length and complexity
- Password reuse
- Common password mistakes
- Password managers
- Multi-Factor Authentication
- Account takeover

- Credential theft
- Password reset scams
- Protecting important accounts

Interactive Tool

Password Strength Checker

Users can test password strength and receive feedback explaining factors such as:

- Password length
- Repeated patterns
- Common password patterns
- Predictability
- Overall password strength

The system should not permanently store user passwords.

Learning Outcome

Users should understand how to create and maintain stronger and more secure accounts.

Module 3 — Phishing & Social Engineering

This module teaches users how attackers manipulate people.

Topics include:

- What is phishing?
- Email phishing
- Smishing
- Vishing
- Social engineering
- Urgency and fear tactics
- Authority impersonation
- Fake login pages
- Suspicious links
- Suspicious attachments
- How to verify suspicious communication

Interactive Tool

Phishing and Message Checker

Users can enter or paste a suspicious message or email into the system.

The tool analyzes indicators such as:

- Urgent language
- Requests for sensitive information
- Suspicious wording
- Impersonation indicators
- Suspicious links
- Social engineering techniques

The platform then provides an educational risk level such as:

- Low Risk
- Medium Risk
- High Risk

It also explains why specific elements may be suspicious.

Learning Outcome

Users should be able to recognize common phishing and social-engineering indicators.

Module 4 — Safe Browsing & Device Hygiene

This module focuses on secure internet and device usage.

Topics include:

- Safe browsing practices
- HTTPS and website awareness
- Suspicious websites
- Fake websites
- Unsafe downloads
- Browser security
- Software updates
- Operating system updates
- Malware prevention
- Application security
- Basic device protection

Interactive Tool

URL Safety Checker

Users can enter a URL for an educational analysis.

The tool can inspect indicators such as:

- HTTPS usage
- Suspicious URL structure
- IP-address-based URLs
- Excessive or misleading subdomains
- Suspicious keywords
- Look-alike patterns

The tool provides an educational risk assessment rather than guaranteeing that a website is completely safe.

Learning Outcome

Users should be able to recognize potentially suspicious websites and maintain better device security practices.

Module 5 — Data Handling & Privacy at Work

This module focuses on protecting personal and organizational information.

Topics include:

- Sensitive information
- Personal data
- Confidential business information
- Data classification concepts
- Safe data sharing
- Cloud storage awareness
- Unauthorized disclosure
- Data leakage
- Privacy
- Secure handling of organizational information
- Sharing information with the correct recipient

Learning Outcome

Users should understand how improper handling of data can create cybersecurity and privacy risks.

Module 6 — Mobile & Remote Work Security

This module focuses on cybersecurity outside the traditional office environment.

Topics include:

- Mobile device security
- Screen locking
- Application permissions
- Software updates
- Public Wi-Fi awareness
- Remote work security
- Laptop security
- Lost or stolen devices
- Secure access practices
- Physical device protection
- Working safely outside the office

Learning Outcome

Users should understand how to maintain cybersecurity while using mobile devices or working remotely.

Module 7 — Security Incident Recognition & Reporting

This module teaches users how to recognize possible security incidents and respond appropriately.

Topics include:

- What is a security incident?
- Suspicious login activity
- Possible account compromise
- Lost or stolen devices
- Accidental data exposure
- Malware warnings
- Unexpected password changes
- Suspicious files
- Unauthorized access indicators
- When to report an incident
- What information should be included in a report

The focus is not on making non-technical users perform advanced forensic investigations.

Instead, the module teaches a simple process:

Recognize → Stop → Preserve Information → Report → Follow Instructions

Learning Outcome

Users should understand when a cybersecurity issue may require reporting and what their role should be during an incident.

Module 8 — Email & Business Communication Security

This module focuses specifically on professional and organizational communication.

Topics include:

- Business email security
- Suspicious attachments
- Suspicious links
- Fake invoices
- Executive impersonation
- Business Email Compromise concepts
- Sender verification
- Payment or bank-detail change requests
- Safe use of CC and BCC
- Misaddressed emails
- Secure communication practices
- Reporting suspicious business emails

Interactive Tool

If time permits, an additional **Email Security Checker** can be implemented to help users identify suspicious characteristics in a business email.

Learning Outcome

Users should understand how to communicate securely and identify common email-based business threats.

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8. Interactive Cybersecurity Tools

The initial version of MilcharSec will contain **4 primary tools**.

Tool 1 — Password Strength Checker

Related Module: Password & Account Security

The user enters a password, and the system evaluates its security characteristics.

The tool provides:

- Password strength level
- Weakness identification
- Improvement recommendations

- Educational explanation

Passwords should be analyzed securely and should not be stored unnecessarily.

Tool 2 — Phishing & Message Checker

Related Module: Phishing & Social Engineering

Users can submit a message or email content for analysis.

The tool identifies potential phishing or social-engineering indicators and provides:

- Risk level
- Detected indicators
- Explanation
- Recommended safe action

Tool 3 — URL Safety Checker

Related Modules: Phishing & Social Engineering and Safe Browsing & Device Hygiene

Users enter a website address for analysis.

The system checks basic characteristics and suspicious indicators and provides an educational assessment.

Possible indicators include:

- HTTPS usage
- URL structure
- Suspicious domains
- IP-based URLs
- Misleading subdomains
- Look-alike patterns

Tool 4 — Cybersecurity Quick Check

This is a platform-wide cybersecurity awareness assessment tool.

The user answers questions related to their security habits and awareness.

Example areas include:

- Password reuse
- MFA usage
- Software updates
- Suspicious links
- Data sharing
- Public Wi-Fi awareness
- Backups
- Device security
- Privacy awareness

The platform generates an overall:

Cybersecurity Awareness Score

For example:

Cybersecurity Awareness Score: 72/100

The system can then identify improvement areas and recommend relevant modules.

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9. Quiz and Assessment System

Each learning module will contain interactive assessments.

The recommended structure is:

Lesson Content

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Interactive Activity

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Mini Quiz

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Scenario-Based Question

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Module Score

Each module can contain approximately **5 to 10 quiz questions**, depending on the amount of content.

Question types may include:

- Multiple-choice questions
- Scenario-based questions
- Identify suspicious behavior
- Identify safe behavior
- Match threats with appropriate responses
- Decision-based security scenarios

The purpose is to test understanding and practical awareness rather than simple memorization.

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10. Practical Scenarios

MilcharSec should include controlled and safe cybersecurity scenarios.

Example Scenario 1 — Suspicious Email

The user receives a simulated email.

They must identify:

- Suspicious sender information
- Urgency tactics
- Suspicious links
- Requests for sensitive information
- Appropriate response

Example Scenario 2 — Password Security

The user is presented with several password examples and must identify safer practices.

Example Scenario 3 — Suspicious Website

The user analyzes a simulated website URL and identifies possible warning signs.

Example Scenario 4 — Security Incident

The user is given a scenario such as:

> An employee receives an unexpected login notification or accidentally shares sensitive information with the wrong recipient.

The user must decide:

- Whether the situation may be a security incident
- What immediate action should be taken
- Whether the incident should be reported

These scenarios help transform theoretical knowledge into practical decision-making.

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11. Scoring and Progress System

MilcharSec will track the user's performance across the platform.

The system may calculate:

- Module completion percentage
- Quiz scores
- Practical scenario scores
- Tool-based activity
- Cybersecurity awareness score
- Overall progress

Example dashboard:

Category	Score
Password & Account Security	85%
Phishing Awareness	62%
Safe Browsing	78%
Data Handling & Privacy	70%
Mobile Security	81%
Incident Recognition	55%

The system can identify weak areas.

For example:

> **Weak Area Identified: Security Incident Recognition**

The platform can recommend:

> **Recommended Next Module: Security Incident Recognition & Reporting**

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12. Personalized Recommendation System

Based on quiz and scenario performance, MilcharSec recommends what the user should learn next.

Example:

> Your score in Phishing & Social Engineering is below your other module scores. It is recommended that you review phishing indicators and complete the practical phishing scenario.

The recommendation system creates a personalized learning path based on performance.

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13. User Dashboard

The learner dashboard should display:

- Overall Cybersecurity Awareness Score
- Modules completed
- Current module
- Quiz performance
- Practical scenario performance
- Tool usage
- Weak areas
- Recommended module
- Recent activity

Example:

Overall Progress: 62%

Cybersecurity Awareness Score: 74/100

Modules Completed: 5/8

Strongest Area: Password & Account Security

Area for Improvement: Phishing Awareness

Recommended Next Module: Phishing & Social Engineering

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14. Administrator Features

The administrator or instructor should be able to:

- Manage users
- Add or edit modules
- Manage lesson content
- Create or edit quizzes
- Manage practical scenarios
- View learner progress
- View quiz results
- Identify common weak areas among users

These features can be implemented based on the available development time.

For the initial MVP, the learner experience should receive higher priority than building an extensive administration system.

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15. Main Features Summary

MilcharSec includes the following major components:

Learning

- 8 cybersecurity awareness modules
- Structured lesson content
- Beginner-friendly explanations
- Workplace-focused cybersecurity education

Interactive Tools

- Password Strength Checker
- Phishing & Message Checker
- URL Safety Checker
- Cybersecurity Quick Check

Assessment

- Module quizzes
- Scenario-based questions

• Practical activities
• Module scores

Progress Tracking

• Module completion
• Quiz performance
• Awareness score
• Weak-area identification

Recommendations

• Personalized module recommendations
• Performance-based learning suggestions

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16. System Architecture

A simple architecture for MilcharSec can be:

Frontend

Web-based user interface for:

• Learning modules
• Dashboard
• Quizzes
• Tools
• User interaction

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Backend

Responsible for:

• Authentication
• User management
• Quiz scoring
• Progress tracking
• Tool processing
• Recommendations

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Database

Stores:

- User information
- Module progress
- Quiz results
- Scores
- Recommendations

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Analysis Layer

Handles the logic for:

- Password strength evaluation
- Phishing indicators
- URL pattern analysis
- Cybersecurity awareness scoring

A modular architecture is recommended so new cybersecurity roles and tools can be added later.

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17. Suggested Technology Stack

The exact stack can depend on the development team's experience.

A possible stack is:

Frontend

- HTML
- CSS
- JavaScript
- React or Next.js
- Tailwind CSS

Backend

- Python FastAPI

or

- Node.js / Express

Database

- PostgreSQL

or, for a smaller MVP:

- Supabase

Deployment

Possible deployment options can include:

- Vercel for the frontend
- Render or similar backend hosting
- Cloud-hosted database services

For a two-week MVP, the team should avoid unnecessary infrastructure complexity.

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18. Future Development

MilcharSec is designed to expand beyond cybersecurity awareness.

Future versions can introduce specialized learning paths.

SOC Analyst Track

Possible features:

- Security log analysis
- Alert triage
- IOC identification
- Incident investigation scenarios

Web Security Track

Possible features:

- Web security fundamentals
- OWASP learning modules
- Security configuration analysis
- Secure development education

Digital Forensics Track

Possible features:

- File integrity analysis
- Hash analysis
- Evidence handling
- Timeline investigation

AI Integration

Future AI features may include:

- AI explanation of cybersecurity concepts
- Personalized learning recommendations
- Guided explanation of security scenarios
- AI-assisted educational feedback

AI should be used to improve explanation and learning rather than being added only as a marketing label.

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19. Project Scope for the Initial Version

The initial MVP will focus on:

8 Learning Modules

1. Cybersecurity Fundamentals
2. Password & Account Security
3. Phishing & Social Engineering
4. Safe Browsing & Device Hygiene
5. Data Handling & Privacy at Work
6. Mobile & Remote Work Security
7. Security Incident Recognition & Reporting
8. Email & Business Communication Security

4 Core Interactive Tools

1. Password Strength Checker
2. Phishing & Message Checker
3. URL Safety Checker
4. Cybersecurity Quick Check

Assessment Features

- Module quizzes
- Scenario-based questions
- Module scoring
- Awareness scoring

Dashboard Features

- Progress tracking
- Module completion
- Quiz results
- Awareness score

• Weak-area identification
• Learning recommendations

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20. Key Innovation and Differentiator

The main differentiator of MilcharSec is not simply the presence of educational modules or cybersecurity tools.

The platform connects them together.

The MilcharSec model is:

Learn

The user studies a cybersecurity topic.

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Practice

The user interacts with a relevant cybersecurity tool or scenario.

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Assess

The user completes quizzes and practical questions.

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Analyze Performance

The system evaluates the user's results.

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Identify Weak Areas

The system determines which cybersecurity topics require improvement.

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Recommend

The user receives a recommendation for the next relevant learning module.

This creates an integrated cybersecurity awareness learning loop.

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21. Expected Outcome

After using MilcharSec, users should have improved awareness of:

- Cybersecurity fundamentals
- Password and account protection
- Phishing and social engineering
- Safe browsing
- Device security
- Data privacy
- Mobile and remote work security
- Security incident recognition
- Secure email and business communication

The expected result is a more informed user who can better recognize common cybersecurity risks and make safer digital decisions.

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22. Conclusion

MilcharSec is a web-based cybersecurity learning, awareness, assessment, and interactive tools platform designed primarily for non-technical users.

The initial version focuses on eight essential cybersecurity modules and four interactive security tools. Users learn cybersecurity concepts, apply their knowledge through tools and practical scenarios, complete quizzes, receive performance scores, and track their cybersecurity awareness progress.

The platform's central concept is:

Learn → Practice → Assess → Track → Improve

MilcharSec is designed as a scalable system. While the initial version focuses on Digital Security Basics and workplace cybersecurity awareness, future versions can expand into specialized cybersecurity roles such as SOC Analysis, Web Security, Incident Response, and Digital Forensics.

The goal is to create a practical and interactive cybersecurity learning environment that goes beyond traditional theoretical awareness training and helps users develop measurable cybersecurity knowledge and safer digital habits.